

Khulna University of Engineering & Technology

Department of Electronics and Communication Engineering

Wireless Power Transfer

– Short-Range Inductive Mobile Charging System

Name	Roll
Md Baized Al Nur	2309033
Abdullah Al Araf Emon	2309048
Yeasir Arafat Siju	2309058

1. Project Title

Wireless Power Transfer – Short-Range Inductive Mobile Charging System.

2. Introduction

Wireless power transfer (WPT) enables electrical energy to move from a source to a load without a direct conductive connection. The proposed project develops a short-range inductive wireless charging prototype for demonstrating this principle at laboratory scale. A 12 V DC supply feeds a KA3525A PWM controller, which produces two alternating gate-drive signals for a pair of IRFZ44N MOSFETs. The MOSFETs excite a center-tapped 6-0-6 turn transmitter coil and establish a high-frequency alternating magnetic field. A nearby receiver coil captures part of this magnetic energy through electromagnetic induction. The induced AC voltage is then rectified by a four-diode 1N5822 Schottky bridge, smoothed by filter capacitors, and regulated to approximately 5 V using a 7805 linear regulator. The regulated output is brought to a USB-C breakout for a small electronic load or compatible low-current mobile-device charging demonstration. The initial design targets operation in the 70–75 kHz region, with the resonant frequency refined from measured coil inductance. The project will also investigate how coil spacing, alignment, orientation, switching frequency, and load affect output voltage and transfer efficiency.

3. Objectives

- To design and make a simple wireless power transfer system using transmitter and receiver coils
- To transfer electrical power from the transmitter coil to the receiver coil through air without direct wire connection
- To study the basic working principle of wireless power transfer and electromagnetic induction
- To calculate the required circuit parameters and component values theoretically before making the circuit
- To select and buy the required electronic components for the project
- To draw and finalize the circuit diagram of the transmitter and receiver sections

4. Proposed System Design

The proposed system is divided into transmitter and receiver sections. The transmitter uses the ATX +12 V rail, input protection, a KA3525A control stage, two IRFZ44N MOSFET switches, and a center-tapped 6-0-6 turn transmitter coil. The receiver uses an 18–20 turn coil, a parallel compensation capacitor, a four-diode Schottky bridge, a DC filter, a 7805 regulator, and a USB-C output interface.

Main features:

- Short-range inductive coupling with an initial coil spacing of about 2–5 mm.
- Adjustable operating point in the approximately 75–80 kHz region, with resonance verified from measured coil inductance.
- Center-tapped 6-0-6 transmitter coil using 24 AWG enameled copper wire and approximately 5 cm diameter.
- 18–20 turn receiver coil of approximately 5 cm diameter.
- High-frequency Schottky rectification and regulated 5 V DC output.
- Low-power laboratory target of about 100–200 mA at 5 V rather than commercial fast-charging performance.

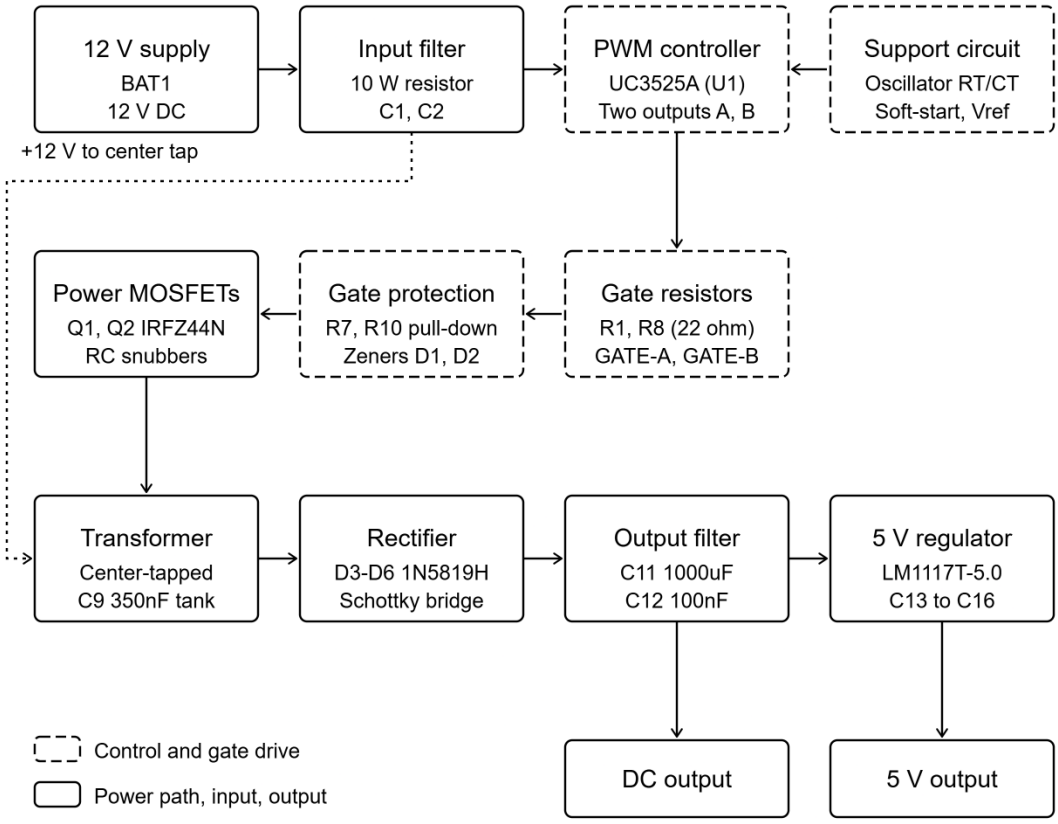


Fig 1: Block diagram of wireless power transfer circuit

Transmitter design:

The KA3525A generates alternating outputs at approximately half the oscillator frequency. The reference design uses RT = 10 kΩ, CT = 1 nF and a 100 Ω discharge/dead-time resistor, giving an oscillator frequency near 150 kHz and alternating gate-drive outputs near 75 kHz. Pins 11 and 14 drive Q1 and Q2 through 22 Ω gate resistors. Each gate has a 10 kΩ pull-down and 15 V zener protection. The TX center tap is supplied through the protected +12 V path; the 300 nF polypropylene capacitor is connected between the two TX coil ends. Series 100 Ω + 1 nF snubbers are placed across the MOSFET drain-source paths.

Receiver design:

The receiver coil is specified as approximately 40.5 µH when measured and is paired with a 100 nF polypropylene capacitor in parallel. The induced AC is rectified by four 1N5822 Schottky diodes, filtered by 1000 µF together with a 100 nF bypass capacitor, and applied to a 7805 regulator. The regulator input/output are supported by 10 µF and 100 nF capacitors. The 5 V node feeds the USB-C VBUS pin.

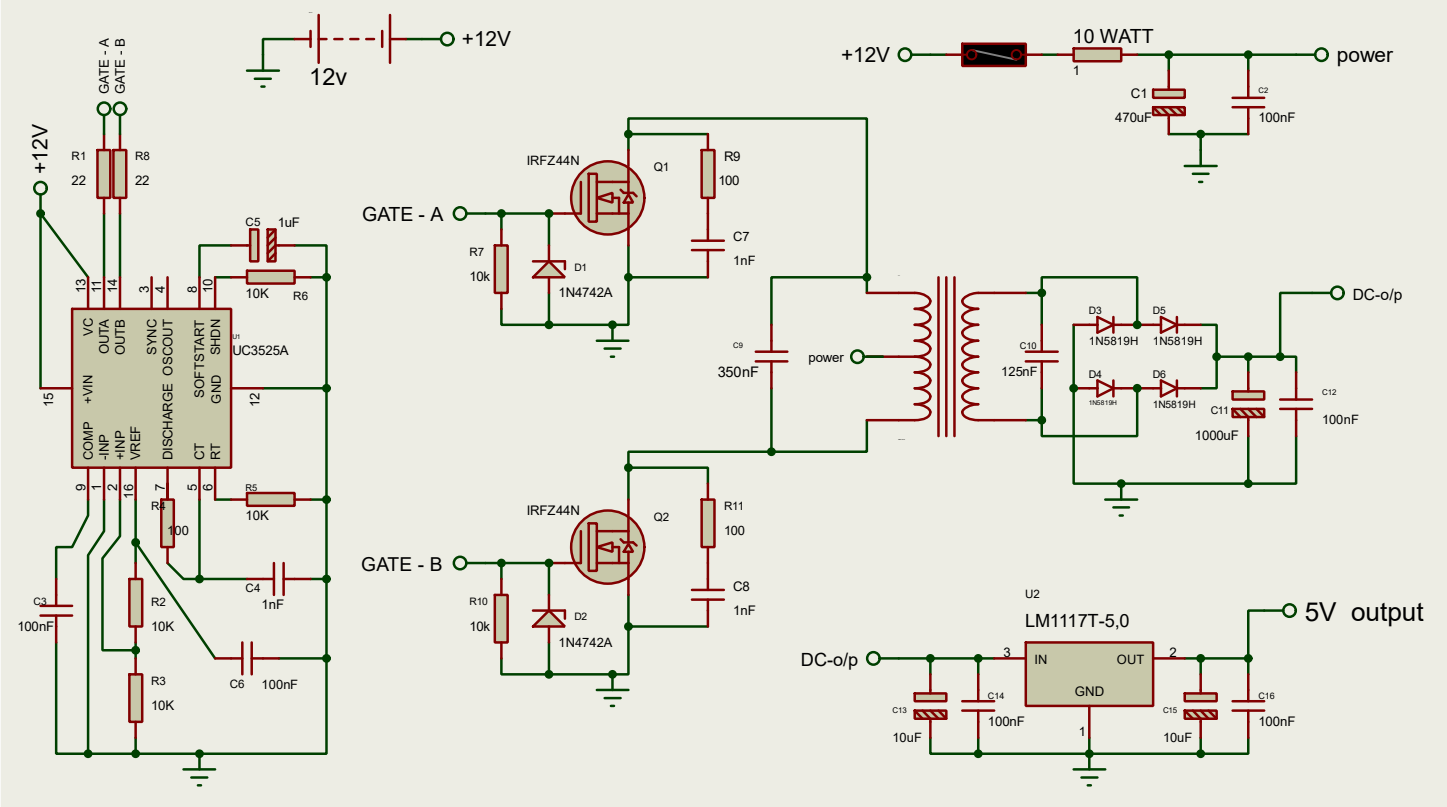


Fig 2: circuit diagram of the wireless power transfer circuit.

5. Proposed Budget for Implementing the Project

Table - 1: The following budget is an estimated local-market planning budget for components:

Sl	Item	Rating	Justification	Price (BDT)
1	KA3525A PWM controller	DIP-16, 8–35 V	Generates alternating control outputs	40
2	IRFZ44N MOSFET	55 V class, 2 pcs	High-frequency power switching	120
3	TX-RX coil + 24 AWG enameled copper wire	18–20 turns, 5 cm	Creating alternating magnetic field and Receives magnetic energy	300
4	Polypropylene film capacitors	100 nF	TX/RX compensation	40
5	1N5822 Schottky diodes	4 pcs, 3 A / 40 V	High-frequency full-wave rectification	40
6	7805 regulator + small heatsink	TO-220	Regulated 5 V output	50
7	Electrolytic/ceramic capacitors	470/1000 µF, 10 µF, 100 nF, 1 µF	Filtering, bypass and soft-start	200
8	Resistors, trimmer, zeners, fuse	Mixed values	Timing, gate protection and input protection	150
9	1 Ω / 5 W resistor	Power resistor	Initial current limiting	10
10	USB-C cable	5 V – 6A	Output interface	80
11	Miscellaneous mounting items	Connectors, spacers, etc.	Mechanical assembly and safe mounting	100
Total estimated project cost (BDT)				**1130**

6. Work Plan

table – 2: week wise work plan

Week	Main Work	Expected Result
1st Week	Research the wireless power-transfer principle, finalize the circuit design, perform resonance and component calculations, review safety requirements, and collect components.	Final design, calculated parameters and components ready.
2nd Week	Construct the 6-0-6 transmitter coil and 18-20 turn receiver coil; assemble and verify the KA3525A controller, timing network and gate-drive section.	Coils and transmitter control electronics assembled.
3rd Week	Build the IRFZ44N MOSFET power stage, verify complementary gate signals, connect the TX tank, and adjust the operating frequency around the required range.	Stable high-frequency transmitter and resonant operation established.
4th Week	Complete the receiver coil, 1N5822 bridge rectifier, filter capacitors and 7805 regulator; obtain and measure the received DC voltage before connecting the load.	Wireless DC output obtained and receiver section verified.
5th Week	Integrate the USB-C output interface, perform load tests, and investigate the effects of coil spacing, alignment, orientation, frequency and load current.	Regulated 5 V output and performance measurements obtained.
6th Week	Optimize coil coupling and operating conditions, analyze voltage/current/power-transfer efficiency, identify limitations, and complete documentation and final demonstration.	Optimized prototype, analyzed results and completed project documentation.

7. Addressing Complex Engineering Problems

The project involves interacting electrical, magnetic, switching and thermal considerations.

Table – 3: following attributes are addressed by the proposed design activities.

Attribute	Tick	Justification
P1 (Depth of knowledge required)	✓	Requires combined knowledge of PWM control, MOSFET switching, resonant circuits, electromagnetic induction, rectification, filtering and voltage regulation.
P2 (Range of conflicting requirements)	✓	The design must balance transfer distance, output power, frequency, resonance, device stress, heat dissipation, circuit size and safe operation.
P3 (Depth of analysis required)	✓	Coil inductance, resonance, switching timing, rectifier loss, regulator dissipation and load-dependent output must be measured and analyzed.
P4 (Familiarity of issues)	✓	The work combines familiar circuit blocks with practical issues such as parasitic effects, coil coupling, alignment sensitivity and high-frequency switching behavior.
P5 (Extent of applicable codes)	—	The proposed laboratory prototype is not designed as a certified commercial charger; therefore, formal product-compliance work is outside the present scope.
P6 (Extent of stakeholder involvement and conflicting requirements)	—	The main requirements are laboratory demonstration and learning objectives, with limited external stakeholder involvement.
P7 (Interdependence)	✓	Transmitter frequency, coil inductance, compensation, coupling, rectifier behavior and regulator input are interdependent and must be optimized together.

8. Addressing Complex Engineering Activities

Attribute	Tick	Justification
A1 (Range of Resources)	✓	Uses a 12 V battery supply, KA3525A PWM controller, IRFZ44N MOSFETs, a center-tapped transformer, 1N5819H Schottky diodes, an LM1117T-5.0 regulator, resistors, capacitors, zener diodes, and simulation (Proteus) and measurement tools.
A2 (Level of Interaction)	✓	Correct operation depends on interaction among the PWM controller, gate drive and protection network, push-pull MOSFET stage, transformer, rectifier, output filter and 5 V regulator.
A3 (Innovation)	_	The project builds on established push-pull converter and PWM control techniques using standard components.
A4 (Consequences for Society and Environment)	✓	The circuit provides a low-power, regulated 5 V supply from a 12 V source, with gate protection, snubbers and filtering to limit switching stress, noise and heat.
A5 (Familiarity)	✓	The project integrates standard circuit blocks with less-familiar high-frequency switching, transformer coupling and rectification behaviour.

9. Expected Outcomes

- A functional low-power wireless power-transfer prototype that transfers energy across a short air gap without a direct conductive connection
- A transmitter producing alternating magnetic excitation in the approximately 70–80 kHz operating region using the KA3525A and two IRFZ44N MOSFETs
- A receiver stage that rectifies the induced AC and provides a regulated output close to 5 V under suitable alignment and loading conditions
- A USB-C output capable of demonstrating approximately 100–200 mA for a small electronic load or compatible low-current charging application
- Experimental evidence showing how spacing, alignment, orientation, frequency and load influence received voltage, output power and transfer efficiency
- Practical understanding of resonance tuning, MOSFET switching, electromagnetic coupling, rectification, regulation, losses and thermal effects